Provenance-Aware Agentic Knowledge Graph Systems for Reliable Multi-Source Grounded AI

Primary Topic: Knowledge Grounding
Secondary Topic: Agentic AI

Retrieval-augmented generation (RAG) [1] has become a foundational technique for grounding large language model (LLM) inference in external knowledge. Production systems have matured substantially: hybrid retrieval combining semantic (dense) and keyword (sparse) search is now standard practice [2], and retrieval is increasingly exposed as a tool within agentic workflows where models decide *when* and *what* to retrieve [3]. Yet despite these advances, two critical gaps remain. First, even state-of-the-art retrieval systems lack structured reasoning substrates capable of multi-hop synthesis across heterogeneous knowledge sources—a gap that graph-based approaches like GraphRAG [4] begin to address but do not fully resolve. Second, current systems provide limited provenance: they can cite documents, but cannot produce fine-grained, verifiable evidence chains linking individual model outputs to specific supporting knowledge, and they lack validation mechanisms that detect and recover from retrieval failures [5], [6].

This project proposes **Provenance-Aware Agentic Knowledge Graph Systems (PA-AKG)**. The core novelty is that PA-AKG builds a *provenance graph* that links every output span to specific KG entities with confidence scores, and deploys validation agents that use this provenance to detect and recover from retrieval failures in real time—no existing system combines structured KG reasoning, adaptive multi-source retrieval, and agent-driven validation in a single grounding substrate [4], [5], [7]. For a multi-hop claim, the provenance graph records which entities, relationships, and source spans contributed at each link, enabling fine-grained verification well beyond document-level citations. Agents dynamically route among dense, sparse, and structured KG pathways based on task context and provenance completeness; validation agents assess evidence sufficiency using provenance signals aligned with observability standards [8], triggering corrective retrieval when grounding is weak or contradictory—extending retrieval-control mechanisms [5], [7] from unstructured documents to structured KG reasoning.

We will prototype PA-AKG on Amazon Bedrock Knowledge Bases [2], using hybrid search as the retrieval layer and validating against the KG provenance backbone. Experiments compare vector-based RAG, hybrid retrieval, GraphRAG-style reasoning, and PA-AKG across **multi-hop QA over heterogeneous knowledge bases** and **claim verification with evidence chain reconstruction**. Phase 1 targets multi-hop QA and claim verification; Phase 2 extends to engineering decision-support. Expected outcomes include the PA-AKG open prototype, provenance-aware benchmarks, and baselines demonstrating that **structured provenance-aware grounding** yields measurable gains in accuracy, verifiability, and failure recovery.

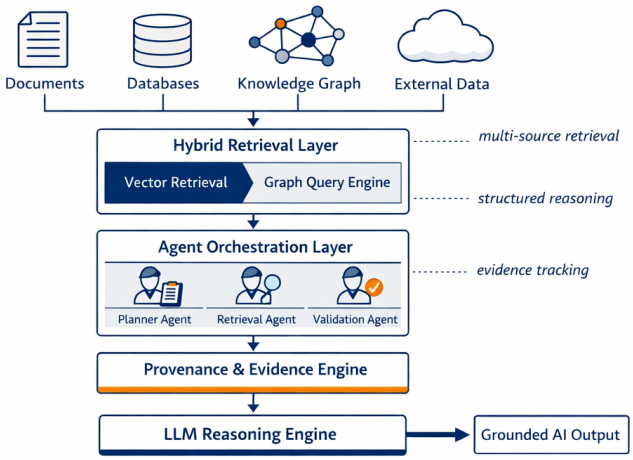

Fig. 1. **PA-AKG Architecture.** Heterogeneous knowledge sources feed a hybrid retrieval layer (vector search + KG queries). An agent orchestration layer coordinates planning, retrieval, and validation agents. A provenance engine records evidence links aligned with observability standards [8] for traceable verification.

References

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.

[2] Amazon Web Services, "Amazon Bedrock knowledge bases now supports hybrid search," AWS Machine Learning Blog, Mar. 2024, accessed 2026-03-10. [Online]. Available: https://aws.amazon.com/blogs/machine-learning/amazon-bedrock-knowledge-bases-now-supports-hybrid-search/

[3] OpenAI, "Function calling guide," OpenAI Developer Documentation, 2025, accessed 2026-03-10. [Online]. Available: https://platform.openai.com/docs/guides/function-calling

[4] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, and J. Larson, "From local to global: A graph RAG approach to query-focused summarization," *arXiv preprint arXiv:2404.16130*, 2024. [Online]. Available: https://arxiv.org/abs/2404.16130

[5] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, "Corrective retrieval augmented generation," *arXiv preprint arXiv:2401.15884*, 2024. [Online]. Available: https://arxiv.org/abs/2401.15884

[6] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.

[7] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. [Online]. Available: https://arxiv.org/abs/2310.11511

[8] OpenTelemetry, "Semantic conventions for generative AI systems," OpenTelemetry Specification, 2024, accessed 2026-03-10. [Online]. Available: https://opentelemetry.io/docs/specs/semconv/gen-ai/